

Scheduled Task/Job: At (T1053.002)

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1 Attack Description in CNL

The attack description of the MITRE ATT&CK Technique “*Scheduled Task/Job: At*” (T1053.002)¹ written in the proposed CNL is shown below.

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1 Tactics: TA0002 Execution, TA0003 Persistence, TA0004 Privilege_Escalation
2 Technique: T1053.002 Scheduled_Task_Job_At
3
4 Background
5 Given process as at_exe with command_line = "at.exe"
6 And process as WMI_Win32_ScheduledJob
7 And process as AnomalousScheduledTask
8 And process as svchost_exe with command_line = "svchost.exe"
9 And process as taskeng_exe with command_line = "taskeng.exe"
10
11 Event 1
12 Given WMI_Win32_ScheduledJob is activated
13 When WMI_Win32_ScheduledJob creates AnomalousScheduledTask
14 Then AnomalousScheduledTask is created
15
16 Event 2
17 Given at_exe is activated
18 When at_exe creates AnomalousScheduledTask
19 Then AnomalousScheduledTask is created
20
21 Event 3
22 Given AnomalousScheduledTask is created And svchost_exe is activated
23 When svchost_exe executes AnomalousScheduledTask
24 Then AnomalousScheduledTask is activated
25
26 Event 4
27 Given AnomalousScheduledTask is created And taskeng_exe is activated
28 When taskeng_exe executes AnomalousScheduledTask
29 Then AnomalousScheduledTask is activated
30
31 Completion
32 Event 3 Xor Event 4
```

¹See <https://attack.mitre.org/techniques/T1053/002/>

2 Attack Model (Petri Net)

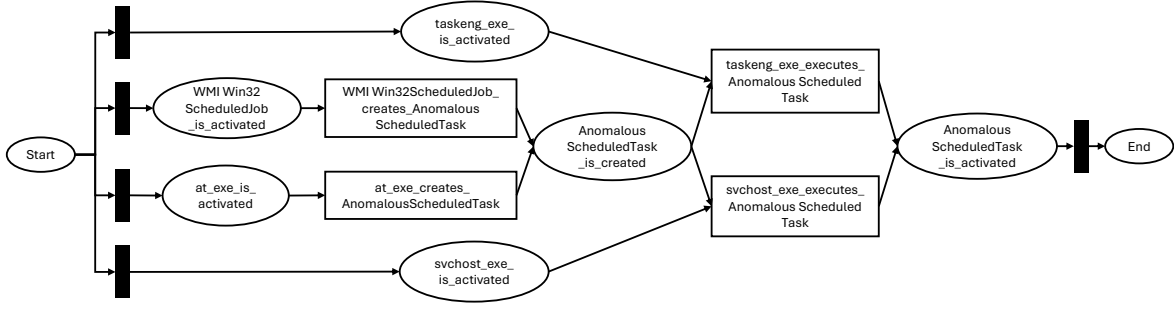


Figure 1: Scheduled Task Job: At Petri Net

3 Runtime and Memory Usage

The translation of this CNL description took 544 ms and allocated 491 KB at peak. The translation pipeline was run on a laptop with Intel Core i7-12650H, 16 GB RAM and running Windows 11.

4 Validation Criteria

The validation process included semi-structured interviews with two domain experts, in which the following validation criteria were discussed: **V1.** the technique description (in CNL) expresses the content of the ATT&CK text; **V2.** the technique description (in CNL) does not contain ambiguities; **V3.** the Technique Model captures the described attack features; **V4.** the Petri Net presents the alternative paths the attack can follow based on the described actions, pre- and post-conditions and assets involved. In the discussion section below, the domain experts are referred to as E1 and E2.

5 Discussion

5.1 V1 & V2

Since the MITRE ATT&CK description for the Technique *Scheduled Task Job: At* contains ambiguous information on how such attack can be detected, the description in CNL was written with the assistance of an expert who provided the correct interpretation of the source description and determined which sequence of events accurately represent the attack. The CNL description provides an explicit expression of the assets participating in the attack, as well as the events that depict a system's behavior when the attack is in action. Particularly, the CNL description shows how the *anomalous process*, as stated in the ATT&CK text, is created and executed on the victim system. In the original text the relation of the *anomalous process* to the scheduled task is left implicit. In the current description the *anomalous process* is described as the actual *scheduled task* created by the *at* scheduler or the *WMI Win32 ScheduledJob*, which is later executed by *svchost.exe* or *taskeng.exe*. E2 highlighted during the interview that this description is more explicit and less ambiguous than the source one.

5.2 V3 & V4

The produced Technique Model successfully captures the attack features as expressed in the CNL description with the appropriate entities, types and composition. The produced Petri Net showcases the alternative ways a system can behave when such attack is executed. Its transitions and places also preserve the action/state verbs and the names of the participating assets as described in the text. The events of alternative processes creating and executing the anomalous scheduled task are directly modeled as different transitions, and ultimately, with their pre- and post-conditions modeled as

intermediate places, they form different execution paths for the attack to follow. During the interview, E2 noted that this particular example of the approach shows that even more complex techniques can be (a) described with the proposed CNL, and (b) translated and modeled into a Petri Net through the implemented pipeline automatically.